



上海交通大学学位论文

理性选择模型：公理化和参数估计

姓 名：程晟

学 号：522120910051

导 师：何滔滔

学 院：安泰经济与管理学院

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**Rational Choice Model:
Characterization and Estimation**

Author: Sheng Cheng

Supervisor: Taotao He

Antai College of Economics and Management
Shanghai Jiao Tong University
Shanghai, P.R.China

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摘 要

本文提出了一种新的随机选择模型，称为选择分支模型（Choice Branching Model, CB），该模型将记忆机制及基于记忆的搜索过程引入消费者的选择行为之中。不同于经典模型（如随机效用模型（RUM）和多项 Logit 模型（MNL））将选择视为在给定备选集合上的一次性决策，我们将选择显式刻画为一个序贯且随机的搜索与分支过程：消费者首先在备选集合中搜索其最偏好的选项；若该选项可得，则立即选择；否则，消费者按照给定的分支规则以一定概率转向其他选项并继续搜索。该过程不断重复，直到找到一个可选的备选项为止。

本模型的一个关键特征是记忆，用于刻画过往搜索结果如何影响后续的分支行为。具体而言，我们假设消费者最多只能记住 k 个已搜索过的备选项，并且分支规则依赖于其记忆状态。我们考察不同记忆水平下模型的性质，使模型能够在“无记忆”和“完全记忆”两种极端情形之间进行刻画。研究表明，随着记忆水平的提高，模型的解释能力呈现出层级递进结构。此外，我们还证明，多项 Logit 模型、马尔可夫链模型以及随机效用模型均可作为该模型的特例。

在此基础上，我们进一步引入一个新的原始变量——反应时间，作为额外的可观测信息。我们发现，反应时间同时反映了偏好结构和记忆水平：记忆能力较低的消费者由于搜索过程更长，通常表现出更长的反应时间；而在面对更偏好的选项时，消费者往往反应更快。这一额外的数据维度在数据有限的情形下显著提升了模型的识别能力。

最后，我们构建了该随机选择模型的极大似然估计方法，并通过基于模拟数据的数值实验验证了该方法的有效性。

关键词：随机选择，搜索和分支，记忆，反应时间

ABSTRACT

This paper proposes a novel stochastic choice model, termed the Choice Branching Model (CB), which incorporates memory and memory-dependent searching into consumer's choice. While classical models such as the random utility model (RUM) and the multinomial logit model (MNL) treat choice as a one-shot selection from a given assortment, we explicitly model choice as the outcome of a sequential and stochastic search-and-branch procedure: A consumer sequentially searches the assortment for the most preferred alternative. If this alternative is available, it is chosen immediately. Otherwise, the consumer stochastically branches to another alternative according to a branching rule and continues the search process. This procedure repeats until an available alternative is found.

A key feature of our model is the memory, which governs the extent to which past search outcomes influence future branching. Specifically, consumers are assumed to recall at most k previously searched alternatives and the branching rule is memory-dependent. We investigate model's property under different memory level, interpolating between 0 memory and full memory branching. We show that varying the memory level induces a hierarchy of models with increasing explanatory range. Besides, we show that widely used models, including MNL, Markov chain models, and RUM, can be obtained as special cases of our model.

We further enrich the model by incorporating a new primitive: response time, as an additional observable. We show that response times are informative about both preference and memory: consumers with lower memory levels tend to exhibit longer response times due to extended search, and choices involving more preferred alternatives are associated with faster responses. This additional dimension of data improves identification, particularly in limited data setting

Finally, we develop a maximum likelihood estimation procedure for the stochastic choice model and evaluate its performance through numerical experiments using synthetic data.

Key words: stochastic choice, searching and branching, memory, response time

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Chapter One: Introduction

1.1 Introduction

Stochastic choice models, which capture the randomness in the consumers' choice behavior, are a core subject in economics, operation management and marketing. A cornerstone of this literature, the random utility model (RUM), was formally introduced by [4] and has since become a foundational paradigm. In RUM, the consumer population is a mixture of heterogeneous consumer types, each of whom has a different rank of choices and chooses the alternative with highest rank in the offered menu. Almost at the same period, [23] introduced the multinomial logit model (MNL) or Luce's model as some literature refers. In MNL, each alternatives are assigned with a weight. When offered an assortment, the consumer are more likely to choose the alternative with relatively high weight.

Ever since then, the RUM and the MNL have been served as two fundamental models in stochastic choice literature. Building upon these two baseline models, many follow-up theoretical works have been done. In these works, they either proposes new models or provide theoretical analysis on existing model's properties. As examples of new-model literature, single-crossing RUM [2], branching-independent RUM [30], (infinite) mixed MNL [27] and nested logit model [21] are directly motivated the baseline models. [7][24] incorporation other behavioral foundations (consideration set and limited attention) on the baseline models. As examples of property-analysis literature, [11] [20] [32] discuss the characterization theorem of RUM. Falmagne [11] first give the characterization theorem of RUM in full choice data set and [20] [32] are the following work investigate the characterization under limited data. [31] discuss the identification of RUM. [14] [10] investigated the choice model from a novel perspective of polytope theory.

Later, after 1960s real-world choice data became more and more popular. As a result, researcher began to explore the application of stochastic choice model. Choice model has been applied in business operation, revenue management, analysis of public transportation and school choice, etc. As examples of application literature, [19] [28] investigated the statistical testing of random utility model. [33] [34] developed a market discovery algorithm to estimate random utility model, which is based on column generation and expectation-maximization technique. McFadden (2001 Nobel laureate in economics for his contribution to the discrete choice model) [26] applied choice model to analyze public transportation system and forecast the demand of transportation. [17] provided a systematic review of

assortment optimization, which is a major application of choice model in revenue management. In assortment optimization literature, the consumers' choice model and the price of each alternatives are given. Researchers' task is to decide which assortment to offer, so that they could maximize the revenue.

1.2 Our Work

We propose a novel stochastic choice model, termed the Choice Branching Model (CB). The CB model is designed to capture consumers' sequential search and choice behavior. Specifically, consumers are assumed to face an assortment of alternatives, with a most-preferred option in mind. They begin by searching the menu to determine whether this preferred alternative is available. If it is, they select it immediately. If not, they proceed to the next most-preferred alternative and repeat the search process. This procedure continues iteratively until a satisfactory alternative is found. Such search-and-choice behavior is pervasive in real-world settings. For instance, consumers often form shopping lists and sequentially search for items on that list. In online shopping environments, it is common for individuals to search for targeted products using search functions, reflecting a structured and goal-directed decision process consistent with the CB framework.

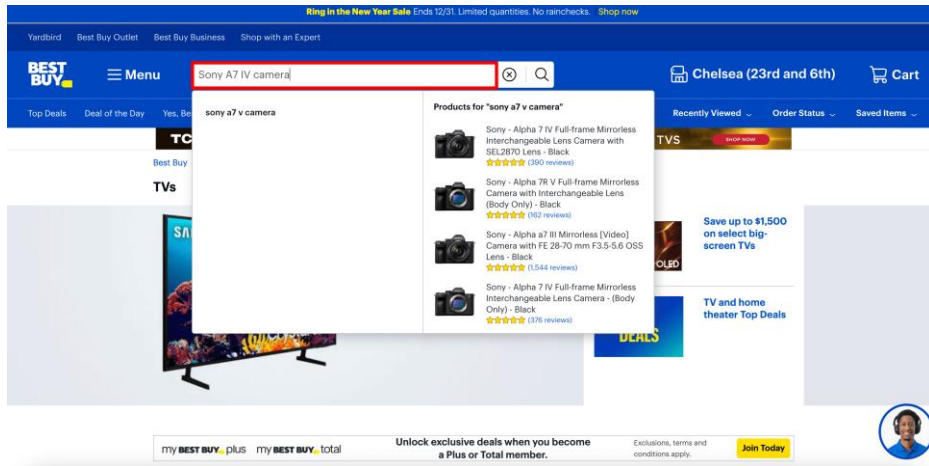


Figure 1: searching bar on online shopping platform

In our model, we introduce a key parameter, referred to as the branching rule, which governs the stochastic component of the decision process. Specifically, after each stage of search, consumers branch to another alternative according to a probability distribution determined by the branching rule. Building on this framework, we further introduce the concept of memory level. We assume that, when branching to subsequent alternatives, consumers can recall at most k previously searched options. As a result, they do not revisit alternatives that

remain in memory, since these have already been searched and found to be unavailable. Accordingly, the branching rule is defined conditional on the consumer's memory set.

Furthermore, extending the static version of the Choice Branching Model, we incorporate an additional primitive: response time. In particular, we assume that the analyst observes not only the alternative chosen from a given assortment, but also the time taken to reach that decision. This additional dimension of data allows the model to capture richer aspects of the underlying search process.

Our main contribution will be the following:

1. First, we develop the *Choice Branching Model* (CB), a novel stochastic choice framework that explicitly incorporates sequential search behavior. We formally define key features of the model, including the memory, the memory updating rule, and the branching rule governing transitions across alternatives. Based on these components, we derive a stochastic choice representation for the CB model.
2. Second, within the static setting, we establish several theoretical results that position CB relative to existing models. We show that widely used models, such as the multinomial logit (MNL), Markov chain models (MCM), and random utility models (RUM), can be obtained as special cases of our framework. We further characterize a hierarchical relationship across different memory levels: CB models with higher memory levels weakly expand the set of choice patterns that can be rationalized by models with lower memory levels. Consequently, the class of CB models indexed by memory level k provides a nested sequence of increasingly rich approximations to RUM in terms of explanatory power. This perspective is closely related to prior work [13], which introduces a conditional independence of irrelevant alternatives (IIA) condition to obtain analogous hierarchical structures. Their approach also motivates our introduction of memory as a behavioral primitive.
3. Third, we enrich the model by introducing a new observable primitive: responding time. It allows us to leverage additional information beyond choice data. We demonstrate that response time can be used to infer memory levels, as lower memory capacity is associated with longer search processes and hence longer response times. Moreover, response time provides a novel source of information for revealed preference, particularly in settings with limited choice data. Even when the set of observed assortments or the number of observations per assortment is small, variation in response times can help reveal preference. At a high level, the model predicts that consumers respond more quickly when choosing more preferred alternatives. Limited data issue has always been a central challenge in decision theory. Our work provides a solution by

incorporating responding times as an additional observable dimension. By exploiting variation in response times, the model is able to identify preference even in environments where traditional identification are hindered.

4. Fourth, we develop a maximum likelihood estimation (MLE) framework for the Choice Branching Model and examine its empirical performance through numerical experiments based on synthetic data. These experiments illustrate the feasibility of the estimation procedure and demonstrate its ability to recover underlying model parameters under controlled settings.

There are three main strands of literature closely related to our work.

1. First, our paper contributes to the stochastic choice literature. We take the MNL [23] and RUM [4] as benchmark frameworks. The Markov chain model (MCM) [3] considers a related form of choice dynamics by modeling transitions across alternatives within a Markovian framework. Our model can be viewed as a generalization of this approach, introducing memory that enrich the state space and allow for history-dependent transitions. Decision tree models [8] adopt a nonparametric approach to capture similar branching structures in choice behavior. Branching-independent RUM [30] imposes independence conditions on transitions to obtain a tractable subclass of RUM with strong identification properties; these conditions are closely related to our notion of set-memory. Similarly, temporal tree models [13] introduce a form of conditional independence (conditional IIA) to regulate model flexibility, generating a hierarchy of models that interpolate between MNL and RUM. This perspective directly motivates our introduction of memory and memory-dependent branching rules.
2. Second, our work relates to the behavioral literature. Our paper is related to literature that incorporates searching into choice. [5] provides a revealed preference framework for stochastic choice with limited consideration, where attention and search constraints shape observed choices. [6] develops a model in which agents sequentially search and stop once an alternative meets a satisficing threshold, emphasizing bounded rationality in stopping behavior. [25] studies environments in which agents iteratively explore alternatives and make decisions based on the outcomes of the search process. Our model differs by explicitly formalizing search as a stochastic branching process with memory dependence. In addition, a growing literature studies response time as an informative behavioral signal, particularly in the context of accuracy and optimal stopping (e.g. [16]). The paper most closely related to ours is [22], which documents a systematic relationship between reported happiness and response time. Our model provides a behavioral foundation for this finding by linking response speed to underlying preferences and search processes.

3. Third, our paper contributes to the literature on choice under limited data. A central challenge in this area is how to test models and identify parameters when observed choice sets or observations are sparse. Existing work [20][32] has developed axiomatic and statistical approaches to RUM with unobserved alternatives and assortments.

1.3 Organization

The remainder of the paper is organized as follows. **Chapter Two: Static Model** introduces the static framework, including the basic primitives and formal definitions of the benchmark models, namely the multinomial logit (MNL) and the random utility model (RUM). We then develop the Choice Branching Model, specifying both the branching process and the memory. **Chapter Three: Theoretical Results in Static Models** presents the main theoretical results for the static model, including the hierarchical relationships across memory levels and identification properties. **Chapter Four: Response Time in Choice Branching** extends the framework by incorporating response time as an additional primitive and derives several results, including the inference of memory levels and the revealed preferences using response time data. **Chapter Five: Estimation and Numerical Experiment** develops the maximum likelihood estimation procedure for the proposed models and reports numerical experiments based on synthetic data to validate the theoretical findings.

Chapter Two: Static Model

In this section, we introduce our static setup including a specification of choice data and stochastic choice rule. Based on the setup, we introduce two fundamental stochastic choice model as our baseline model: the random utility model and the multinomial logit model. Motivated by these two models, we extended our paper to the static version of choice branching model (with limited memory).

Note that various stochastic choice models has been proposed to capture different choice behavior, we refer readers to the book [29], which provide an excellent review of stochastic choice theory. Besides, notations in our paper mostly follows this book.

2.1 Basic Setup

We specify the choice data that we can collect. Let X be a finite set of alternatives. Let $\mathcal{X} := 2^X / \{\emptyset\}$ be the set of assortments. We assume that when we offer an arbitrary assortment $A \in \mathcal{X}$ to the consumers, we are able to observe the their choice from that assortment $s \in A$. We denote the set of possible choice observation as $I := \{(x, A) : A \in \mathcal{X}, x \in A\}$. A choice dataset contains a collection of choice observation, and each choice observation must be in I . We notice that only the number of choice observation $(x, A) \in I$ matters in a choice dataset. Thus, we denote a choice dataset as $N \in \mathbb{Z}^{|I|}$, where $N(x, A)$ record the number of choice observation (x, A) appears in the choice dataset. Next, we introduce the model framework that we will follow through out the paper.

Definition 1. $p: I \rightarrow \mathbb{R}_+$ is a *stochastic choice rule* if it satisfies following:

- $\sum_{x \in A} p(x, A) = 1$ for all $A \in \mathcal{X}$.

Intuitively, $p(x, A)$ represents the probability of choosing alternative s from assortment A . When each assortment is offered to consumer for infinite times, we can obtain their stochastic choice rule. Note that due to the limited size of data, we can only obtain choice frequencies in application. The gap between frequency and probability, motivates the research on statistical testing and estimation on choice models.

2.2 Baseline Models

In this section, we will introduce two baseline models: the random utility model (RUM) and the multinomial logit model (MNL). A choice model is based on the framework of stochastic choice rule. A choice model introduce model parameters and assumption on consumers' behavior and structuralize the stochastic choice rule p using a functional form.

First, we introduce the random utility model. In RUM, each consumer has a linear preference over X (a permutation of X) and there are $|X|!$ types of consumer in total. Consumer population is a distribution of $|X|!$ types of consumer. The stochastic choice rule for the consumer population is an aggregation of choice from each consumer. Formally, let $\succ$ denote a strict linear order over X and $\mathcal{L}(X)$ denote the set of linear orders over X . Let $\mu \in \Delta(\mathcal{L}(X))$ denote a probability distribution over linear orders of X . Consider a set of linear orders,

$$L(x, A) := \{\succ \in \mathcal{L}(X) : x \succ t, \forall t \in A \setminus x\}.$$

Intuitively, $L(x, A)$ is the set of linear orders, where x is the best among A . We give the formal definition of RUM as the following.

Definition 2. A stochastic choice rule p has a **random utility** representation if there exists a $\mu \in \Delta(\mathcal{L}(X))$ such that

$$p(x, A) = \sum_{\succ \in L(x, A)} \mu(\succ), \forall (x, A) \in I.$$

Next, we introduce the multinomial logit model or Luce model referred in some literature. In MNL model, a weight is assigned to each alternative and the stochastic choice rule are structuralized as a fractional function. Let $w: X \rightarrow \mathbb{R}$ denote a weight function over alternatives. We give the formal definition of MNL as the following.

Definition 3. A stochastic choice rule p has a **multinomial logit** representation if there exists a weight function w such that

$$p(x, A) = \frac{w(x)}{\sum_{a \in A} w(a)}, \forall (x, A) \in I.$$

2.3 Finite Mixed Multinomial Logit Model

In this subsection, we will introduce a group of novel models, which is extended from the baseline model. Models include finite mixed multinomial logit model and choice branching

model.

In finite mixed multinomial logit model, we assume the consumer population consists of $k \in \mathbb{N}_{++}$ heterogeneous consumer types. Each consumer type admits a multinomial logit representation. Let $\lambda \in \Delta(\{1, 2, \dots, k\})$ denote a probability distribution over types. We give the formal definition of finite mixed multinomial logit model as the following:

Definition 4. A stochastic choice rule has a ***k* mixed multinomial logit** representation if there exists a $\lambda \in \Delta(\{1, 2, \dots, k\})$ and a collection of weight function $(w_i)_{i=1}^k$ such that

$$p(x, A) = \sum_{i=1}^k \lambda(i) \cdot \frac{w_i(x)}{\sum_{a \in A} w_i(a)}.$$

For abbreviation, we use k-MNL to refer the k mixed multinomial logit model. Trivially, MNL is a special case of k-MNL, as in MNL we assume there is only one consumer type in the population. Note that similar idea of mixed MNL model has been discussed before. For example, [27] discussed infinite mixed MNL model, which does not put a limit on the number of heterogeneous types in population. The major difference is that our model assumes a finite number k of consumer types.

2.4 Choice Branching Model

We introduce the second model: choice braching model with limited memory. To make the idea more natural, we will start with an real-world example.

Example 1. A consumer walked into the grocery shop to buy drinks. The first alternative he targeted to buy is Pepsi. He searched for Pepsi on the shelf but only to find Pepsi is out of storage. Then, his targeted alternative transit to Sprite. He repeated the searching and found nothing. After rounds of choice branching and searching, he ended up with Fanta, which is offered on the shelf.

The example describes a choice branching process: consumer faces a choice problem with a targeted alternative x_1 in his mind and check if the alternative is offered in the assortment. If it is, consumer chooses x_1 and terminates the choice problem. Otherwise, consumer transit to the next targeted alternatives following a certain branching probability. Then he check whether the second alternative is offered. Consumer repeat this process until the targeted alternative is in the assortment.

Moreover, we introduce a novel notion: *limited memory*. In choice branching model with k-set memory, we assume that consumers can only recall the set (without order) of k alternatives that they have searched. In choice branching model with k-sequence memory, assume that consumers can only recall the sequence (with order) of k alternatives that they have searched. In next paragraph, we will formalize this idea.

Let $\pi: \{1, 2, \dots, |S|\} \rightarrow S$ denote a nonrepetitive sequence. Let E_π denote the element in the permutation π . Let $|\pi|$ denote the cardinality of the sequence, i.e., $|S|$. In the choice branching process with k memory level, a **memory** ω is a nonrepetitive sequence such that $|\omega| \leq k$. Let $\Omega_k := \{\omega: E_\omega \subseteq X, |\omega| \leq k\}$ denote the **memory space**. The memory ω means that consumers just searched E_ω with an order recorded by ω . For example, $\omega(|\omega|)$ is the latest alternative that consumers have searched.

A stochastic process is a indexed family of random variable:

$$\{X_t: t \in T\}.$$

The **choice branching process** with k-level memory is a discrete-time stochastic process with memory space Ω_k :

$$\{X_t \in \Omega_k, t \in T = \{0, 1, 2, \dots\}\}.$$

The choice branching process with k-level memory works like this. The initial memory of consumer is $\emptyset$, i.e., $\Pr(X_0 = \emptyset) = 1$. At each memory, consumer branches to the next alternative x^* following a subjective choice branching rule and updates the memory follows the rules below.

- If $|\omega| < k$, the new state ω^* is where $\omega^*(i) = \omega(i) \quad \forall i \leq |\omega|$ and $\omega^*(|\omega| + 1) = x^*$.
- If $|\omega| = k$, the new state is $\omega^*: \{1, 2, \dots, |\omega|\} \rightarrow S \cup x^*/\omega(1)$, where $\omega^*(i) = \omega(i + 1) \quad \forall i \leq |\omega| - 1$ and $\omega^*(|\omega|) = x^*$.

In **Definition 5** and **Definition 6**, we define two choice branching rules. $\psi_k(\omega, x)$ and $\phi_k(\omega, x)$ means the probability of branching to the alternative x conditional on the memory ω . Condition 1 means that consumers will not branch to an alternative in their memory. It is reasonable, as alternatives in memory have been searched and are not available. Condition 2 is the sum-up-to-one condition of branching probability, which makes ρ_k, γ_k valid probability measures. Condition 3 is exclusive to the k-set choice branching rule. It means that the consumers can recall previous k searched alternatives but cannot recall the order of searching. Thus, we replace the **Definition 6** with an abbreviated form **Definition 6***.

Definition 5: A function $\psi_k: \Omega_k \times X \rightarrow \mathbb{R}_+$ is a ***k*-sequence choice branching rule** if it satisfies the following:

1. $\psi_k(\omega, x) = 0$, for all $\omega \in \Omega_k$ and $x \in E_\omega$;
2. $\sum_{x \in X \setminus E_\omega} \psi_k(\omega, x) = 1$, for all $\omega \in \Omega_k$.

Definition 6: A function $\phi_k: \Omega_k \times X \rightarrow \mathbb{R}_+$ is a ***k*-set choice branching rule** if it satisfies the following:

1. $\phi_k(\omega, x) = 0$, for all $\omega \in \Omega_k$ and $x \in E_\omega$;
2. $\sum_{x \in X \setminus E_\omega} \phi_k(\omega, x) = 1$, for all $\omega \in \Omega_k$;
3. $\phi_k(\omega, x) = \phi_k(\omega', x)$, for any two permutation ω_S, ω'_S , where $S \in \mathcal{X}, |S| \leq k$, and $x \in X \setminus E_\omega$.

Definition 6*: Let $\mathcal{S} := \{S: S \in \mathcal{X}, |S| \leq k\}$. A function $\phi_k: \mathcal{S} \times X \rightarrow \mathbb{R}_+$ is a ***k*-set choice branching rule** if it satisfies the following:

1. $\phi_k(S, x) = 0$, for all $S \in \mathcal{S}$ and $x \in S$;
2. $\sum_{x \in X \setminus S} \phi_k(S, x) = 1$, for all $S \in \mathcal{S}$.

Based on the state updating rule and choice branching, we can further specify the **memory transition matrix** in the following proposition.

Proposition 1.

Given a choice branching process with *k*-memory level and a *k*-sequence choice branching rule ρ_k or *k*-set choice branching rule γ_k , the probability from state ω to state ω' $P(\omega, \omega')$ is the following

1. If $|\omega| = |\omega'| - 1 < k$, $\omega(i) = \omega'(i)$ for all $i \leq |\omega|$ and $\omega'(|\omega'|) \notin E_\omega$, **OR** If $|\omega| = |\omega'| = k$, $\omega(i) = \omega'(i - 1)$ for all $2 \leq i \leq k$ and $\omega'(k) \notin E_\omega$,

$$P(\omega, \omega') = \rho_k(\omega, \omega'(|\omega'|)) \text{ or } P(\omega, \omega') = \gamma_k(\omega, \omega'(|\omega'|));$$

2. Otherwise,

$$P(\omega, \omega') = 0.$$

Finally, to formalize the definition of choice branching model, we derive the stochastic choice rule p from the choice branching process and choice branching rule ρ_k and γ_k . For an assortment $A \in \mathcal{X}$ and an alternative $x \in A$, let $F(x, A)$ be a set of choice branching path that,

$$F(x, A) := \left\{ \omega = \{\omega_t\}_{t \in \{0,1,\dots,T\}} \left| \begin{array}{l} T \in \mathbb{Z}_{++}; \\ \omega_0 = \emptyset, \omega_T(|\omega_T|) = s; \\ \forall a \in A; a \notin E_{\omega_1}, \dots, E_{\omega_{T-1}} \end{array} \right. \right\}$$

$F(s, T)$ records the events such that s is the last alternative and $\forall x \in T$ are not trasversed during the choice branching process. Then, based on the model we describe above, a p that admits choice branching representation can be expressed as the following.

$$\begin{aligned} p(x, A) &= \sum_{\omega \in F(x, A)} \Pr(X_0 = \omega_0, \dots, X_T = \omega_T) \\ &= \sum_{\omega \in F(x, A)} 1 \cdot P(\omega_0, \omega_1) \cdot P(\omega_1, \omega_2) \cdot \dots \cdot P(\omega_{T-1}, \omega_T) \end{aligned}$$

For the k -sequence branching rule ψ_k ,

$$p(x, A) = \sum_{\omega \in F(x, A)} \psi_k(\omega_0, \omega_1(|\omega_1|)) \cdot \psi_k(\omega_1, \omega_2(|\omega_2|)) \cdot \dots \cdot \psi_k(\omega_{T-1}, \omega_T(|\omega_T|)) \quad (1)$$

For the k -set branching rule ϕ_k ,

$$p(x, A) = \sum_{\omega \in F(x, A)} \phi_k(E_{\omega_0}, \omega_1(|\omega_1|)) \cdot \phi_k(E_{\omega_1}, \omega_2(|\omega_2|)) \cdot \dots \cdot \phi_k(E_{\omega_{T-1}}, \omega_T(|\omega_T|)) \quad (2)$$

Hence, we have the following definition.

Definition 7.

1. A stochastic choicerule p has a **choice branching with k -sequence memory (k -SQCB)** representation if there exists a k -sequence branching rule ρ_k such that p satisfies Equation (1).
2. A stochastic choicerule p has a **choice branching with k -set memory (k -STCB)** representation if there exists a k -set branching rule γ_k such that p satisfies Equation (2).

Figure 2 and figure are illustrations of choice branching process. In **Figure 2**, $X = \{a, b, c, d\}$ and a menu $\{a, b\}$ is offered to a consumer who admits choice branching process with 2-memory level. Blue dot means the termination of the choice branching process (consumer will choose the corresponding alternative at the termination). In **Figure 3**, $X = \{a, b, c\}$ and a menu $\{a, b\}$ is offered to a consumer who admits choice branching process with 0-memory level. Note that in **Figure 2**, the branching process will reach the termination for all circumstances. However, in **Figure 3** the branching process will continue

infinitely in some circumstances. Suppose that the memory level is k and menu size is t , we notice that when $k + t < |X|$, infinite branching will happen.

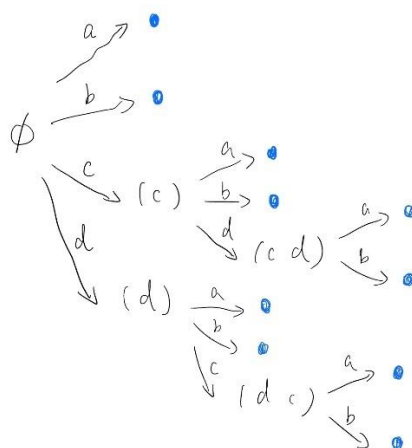


Figure 2: illustration of the choice branching with 2-memory level

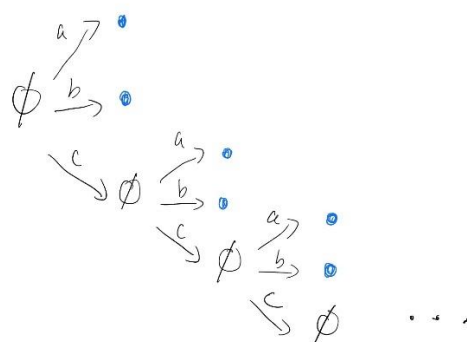


Figure 3: illustration of the choice branching with 0-memory level

Here we add a remark to emphasize some important points in this model.

Remark 1.

- Note that the choice branching process is unobservable. That is we cannot see the alternatives that consumers traversed. We can only see the their final choice.
- *Memory affects the choice branching process from two perspectives. First, it determines the set of alternatives that a consumer can branch at each states. Intuitively, if consumers can recall S as the set of searched alternatives, they will not waste their energy on S . However, there might exists repetition in a choice branching process, as the consumers further proceeds branching they forgot some searched alternatives. Second, the choice branching probability are conditional on the memory. Independence*

is implicitly included, as choice branching are independent on the history before the memory.

Our choice branching model is related to some other choice models. The branching-independent RUM [30] discussed such choice branching process and introduce an additional branching-independent assumption on RUM to derive a identifiable model. The Markov Chain model [3] also models the branching behavior based on the Markov chain framework. However, Markov chain model is a special case of our model with 1-level memory. The temporal tree model [13] discuss the level of independence in the choice branching process which motivates the notion of memory level in our paper. However, we develop the model in a different way from them which aligns more closely with behavioral foundation. Moreover, we will introduce response time as a new primitive in this model, which will serve as an extension.

Chapter Three: Theoretical Results in Static Models

3.1 Hierarchical Connection with Models

In this subsection, we will discuss the connection between choice branching model with some other models, including MNL, RUM, Branching-independent RUM (BIRUM) and Markov Chain model (MCM).

For a choice model that structuralize the stochastic choice rule, we are interested in its explanatory range. In other word, we want to know what the collection stochastic choice rule that can be explained by model exactly is. To formalize, we introduce the notion of explanatory range. Given a choice model **Model**, the explanatory range is a set $\mathcal{D}_{Model}$ such that

$$\mathcal{D}_{Model} := \{p \in \mathbb{R}^I \mid p \text{ admits a } \mathbf{Model} \text{ representation}\}$$

Then, let $\mathcal{D}_{MNL}, \mathcal{D}_{RUM}, \mathcal{D}_{BIRUM}, \mathcal{D}_{MCM}, \mathcal{D}_{kSQCB}, \mathcal{D}_{kSTCB}$ denote the explanatory range of these models respectively. In the next proposition, we discuss the connection between different models from the perspective of explanatory range.

Proposition 2.

1. p has a MNL rep. if and only if p has a choice branching with 0-sequence(set) memory rep.. That is $\mathcal{D}_{MNL} = \mathcal{D}_{0-SQCB} = \mathcal{D}_{0-STCB}$.
2. p has a MCM rep. if and only if p has a choice branching with 1-sequence(set) memory rep.. That is $\mathcal{D}_{MCM} = \mathcal{D}_{1-SQCB} = \mathcal{D}_{1-STCB}$.
3. The following three statements are equivalent:
 p has a RUM rep.;
 p has a BIRUM rep.;
 p has a choice branching with full memory choice branching rep..
That is $\mathcal{D}_{RUM} = \mathcal{D}_{BIRUM} = \mathcal{D}_{|X|-SQCB} = \mathcal{D}_{|X|-STCB}$.
4. For any k , $\mathcal{D}_{k-SQCB} \subseteq \mathcal{D}_{k-STCB}$.
5. For any $0 \leq k \leq |X| - 1$, if p has a choice branching with k -sequence(set) memory rep., then p has a choice branching with $k+1$ -sequence(set) memory rep.. That is $\mathcal{D}_{k-SQCB} \subseteq \mathcal{D}_{k+1-SQCB}, \mathcal{D}_{k-STCB} \subseteq \mathcal{D}_{k+1-STCB}$

Item 2 4 and 5 are trivial. We provide the proof of item 1 and 3.

Proof.

Proof of item 1. Since 0-sequence memory and 0-set memory are equivalent, it suffices to

show one of them. Suppose p has 0-SQCB(0-STCB) representation, let $\rho(\emptyset, x) \forall x \in X$ be the branching probability. Write $\rho(x)$ as abbreviation of $\rho(\emptyset, x)$. Then, $\forall (x, A) \in I$,

$$\begin{aligned} p(x, A) &= \rho(x) + \sum_{a \in X \setminus A} \rho(a) \rho(x) + \sum_{a \in X \setminus A} \sum_{b \in X \setminus A} \rho(a) \rho(b) \rho(x) + \dots \\ &= \rho(x) \left(1 + \sum_{a \in X \setminus A} \rho(a) + \sum_{a \in X \setminus A} \sum_{b \in X \setminus A} \rho(a) \rho(b) + \dots \right) \\ &= \rho(x) \left[1 + \sum_{a \in X \setminus A} \rho(a) \left(1 + \sum_{a \in X \setminus A} \rho(a) + \sum_{a \in X \setminus A} \sum_{b \in X \setminus A} \rho(a) \rho(b) + \dots \right) \right] \end{aligned}$$

For simplicity, let S denote $1 + \sum_{a \in X \setminus A} \rho(a) + \sum_{a \in X \setminus A} \sum_{b \in X \setminus A} \rho(a) \rho(b) + \dots$. Then, by equality of line 2 and line 3, we have

$$\rho(x) \cdot S = \rho(x) \left(1 + \sum_{a \in X \setminus A} \rho(a) \cdot S \right).$$

Then,

$$S = \frac{1}{\sum_{a \in A} \rho(a)}.$$

Then,

$$p(x, A) = \rho(x) \cdot S = \frac{\rho(x)}{\sum_{a \in A} \rho(a)}.$$

Therefore, p has a MNL representation with weight function being ρ . Conversely, suppose p has a MNL representation with w being the weight function. Let $\rho(\emptyset, x) := w(x) \forall x \in X$. It is easy to show that

$$\frac{w(x)}{\sum_{a \in A} w(a)} = w(x) + \sum_{a \in X \setminus A} w(a) w(x) + \sum_{a \in X \setminus A} \sum_{b \in X \setminus A} w(a) w(b) w(x) + \dots$$

Therefore, p admits a choice branching with 0-memory rep.

Proof of Item 3. [30] has proved that statement 1 and statement 2 are equivalent. Need to show 1 and 3 are equivalent. We provide a proof sketch. Suppose p admits choice branching with full memory. Let ρ be the branching probability. For any linear order $\succ \in \mathcal{L}(X)$ such that $x_1 \succ x_2 \succ \dots \succ x_{|X|}$, let

$$\mu(\succ) = \rho(\emptyset, x_1) \cdot \rho((x_1), x_2) \cdot \rho((x_1, x_2), x_3) \cdot \dots \cdot \rho((x_1, x_2, \dots, x_{|X|-1}), x_{|X|})$$

Next we show that p admits a RUM rep. with μ being the distribution of preference.

Q.E.D

In **Proposition 2.**, item 1&2&3 shows that MNL, MCM, RUM are special cases of the choice branching model. Item 4 shows the connection between sequence memory and set memory. Item 5 shows the connection between CB with different memory level. Together, we see that CB serves as a hierarchical inner approximation of RUM in terms of explanatory range. To better illustrate, we provide an example and a figure. **Example 2.** shows the connection between 0-memory CB and MNL.

Example 2. Let $X = \{a, b, c\}, A = \{a, b\}$. Suppose a admits 0-CB with branching rule ρ . Write $\rho(x)$ as abbreviation of $\rho(\emptyset, x)$. Then,

$$\begin{aligned} p(a, A) &= \rho(a) + \rho(c)\rho(a) + \rho(c)^2\rho(a) + \dots \\ &= \rho(a) \cdot \lim_{t \rightarrow \infty} \frac{1 - \rho(c)^t}{1 - \rho(c)} = \frac{\rho(a)}{\rho(a) + \rho(b)}. \end{aligned}$$

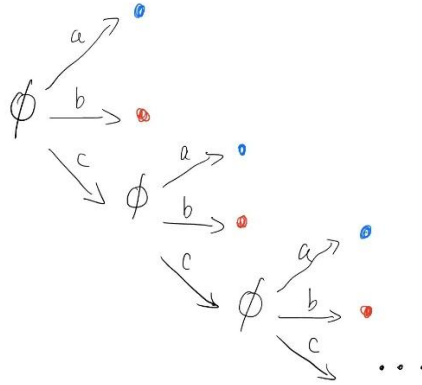


Figure 4: illustration of **Example 2**

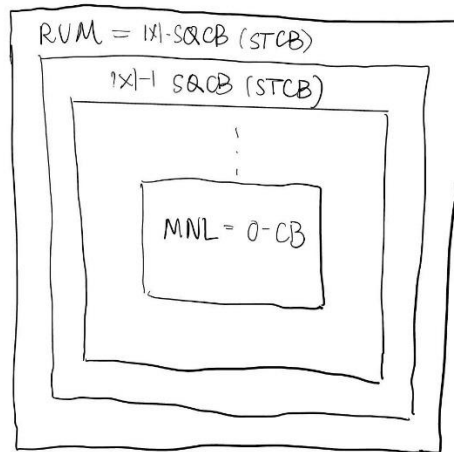


Figure 5: illustration of **Proposition 2.** item 4, the hierarchical connection of CB

There is a general tradeoff between structure and explanatory range. A more structural model with simple model representation (e.g. MNL) normally fail to explain many choice. Thus, estimating such model with choice data may lead to large mean square error. A less structural model with heavy model representation (e.g.) can approximate a large range of choice data. In contrast, estimating such model will have heavy computation burden but small mean square error. However, note that larger explanatory range is not necessarily better, as there might exist overfitting problem. For example, at the most extreme case, a model with zero structure that explains any choice data will have no prediction power. In our choice branching model, higher memory level implies less structure and larger explanatory range. Therefore, we offer flexibility for model user to balance between the two ends.

From **Proposition 2**. item 1&2&3, we see that CB with 0,1 and full memory level covers exactly explanatory range of MNL, MCM and RUM respectively. We can go one step further to the parameter space. There are one-to-one correspondences between parameters of CB and parameters of MNL, MCM, RUM. Such correspondences can be examined by looking at the number of parameters.

<i>Model</i>	<i>Number of parameters</i>
<i>MNL</i>	$ X $
<i>MCM</i>	$ X ^2$
<i>RUM</i>	$ X !$
<i>CB with 0 memory</i>	$ X $
<i>CB with 1 memory</i>	$ X ^2$
<i>CB with full memory</i>	$ X !$

Table 1. number of parameters of models

For MCM, correspondence is trivial, as 1-SQ(ST)CB and MCM follows the same definition. For MNL and RUM, we use the following proposition to formalize our idea

Proposition 3

1. For any p that admits MNL rep. w , p also admits a 0-SQ(ST)CB rep. ψ_0 (or ϕ_0 equivalently) such that w and ψ_0 (or ϕ_0) has an one-to-one correspondence.
2. For any p that admits RUM rep. μ , p also admits a $|X|$ -SQ(ST)CB rep. $\psi_{|X|}$ (or $\phi_{|X|}$) such that w and $\psi_{|X|}$ (or $\phi_{|X|}$) has an one-to-one correspondence.
3. For any p that admits BIRUM rep. μ' , p also admits a **unique** $|X|$ -STCB rep. $\phi_{|X|}$ such that μ' and $\phi_{|X|}$ has an one-to-one correspondence.

3.2 Identification

In this subsection, we will discuss the identification of CB model. That is given a choice data p , what is the parameters of the model $\rho_k \gamma_k$. Furthermore we can ask the following question: Are the parameters pinned down uniquely? If the uniqueness fails, the there are some parameters $\rho_k \gamma_k, \rho'_k \gamma'_k$ that represents the same p .

We examine the identification of two extreme cases: CB with 0 memory and CB with full memory. CB with 0 memory is identifiable. When universal assortment is X offered, ρ_0 or γ_0 equivalently can be identified as the following.

$$\rho_0(x) = p(x, X) \forall x \in X.$$

However, RUM is unidentifiable. Such problem is famous and discussed by many literature [31]. Fishburn [15] provided an excellent example (See **Example 3**) to illustrate the unidentification. This problem is natural since the choice data space $R^{|I|}$ has much smaller dimensions than parameter space $R^{|\Delta(\mathcal{L}(S))|}$: $|I| = o(2^{|X|}) < |\Delta(\mathcal{L}(S))| = o(|X|!)$.

Example 3. Let $X = \{a, b, c, d\}$. The following three RUMs μ_1, μ_2, μ_3 represent the same p .

$$\begin{aligned} \mu_1(a > b > c > d) &= \mu_1(b > a > d > c) = 0.5; \\ \mu_2(a > b > d > c) &= \mu_2(b > a > c > d) = 0.5; \\ \mu_3(a > b > c > d) &= \mu_3(b > a > d > c) \\ &= \mu_3(a > b > d > c) = \mu_3(b > a > c > d) = 0.25. \end{aligned}$$

Chapter Four: Response Time in Choice Branching

4.1 New Primitive: Response Time

In this section, we introduce a new primitive: response time. That is when an assortment is offered to a consumer, we can not only observe their choice but also their repending time to make the choice. We assume in choice branching model, each searching takes one unit of time. Suppose a choice branching process takes a total of n searches to terminate. Then a extended choice data is (x, A, n) representing consumer chooses x from assortment A using n units of time to respond. Let $I^* := \{(x, A, n): A \in \mathcal{X} \ x \in A \ n \in \mathbb{Z}_{++}\}$ denote the extended possible choice observation set.

Definition 8 A function $q: I^* \rightarrow \mathbb{R}_{++}$ is a **time-extended stochastic choice rule** if it satisfies

- $\sum_{x \in A} \sum_{n \in \mathbb{Z}_{++}} q(x, A, n) = 1 \ \forall A \in \mathcal{X}.$

We denote $\sum_{n \in \mathbb{Z}_{++}} q(x, A, n)$ as $q(x, A)$. $q(x, A, n)$ means the probability of choosing x from A in n units of time. Moreover we define average response time of choice observation (x, A) as $\tau(x, A) := \sum_{n \in \mathbb{Z}_{++}} n \cdot q(x, A, n)$. Next we define the time-sensitive choice branching model. Let

$$F'(x, A, n) := \left\{ \omega = \{\omega_t\}_{t \in \{0, 1, \dots, T\}} \left| \begin{array}{l} T = n; \\ \omega_0 = \emptyset, \ \omega_T(|\omega_T|) = s; \\ \forall a \in A; \ a \notin E_{\omega_1}, \dots E_{\omega_{T-1}} \end{array} \right. \right\}.$$

Then based on above description, for k -sequence choice branching rule

$$q(x, A, n) = \sum_{\omega \in F'(x, A, n)} \psi_k(\omega_0, \omega_1(|\omega_1|)) \cdot \psi_k(\omega_1, \omega_2(|\omega_2|)) \cdot \dots \cdot \psi_k(\omega_{T-1}, \omega_T(|\omega_T|)). \quad (3)$$

For k -set choice branching rule

$$q(x, A, n) = \sum_{\omega \in F'(x, A, n)} \phi_k(\omega_0, \omega_1(|\omega_1|)) \cdot \phi_k(\omega_1, \omega_2(|\omega_2|)) \cdot \dots \cdot \phi_k(\omega_{T-1}, \omega_T(|\omega_T|)) \quad (4)$$

Definition 9. A time-extended stochastic choice rule q has a **time-extended k -sequence**

choice branching representation (k -TSQCB) if there exists a ψ_k such that q satisfies (3).

Definition 10. *A time-extended stochastic choice rule q has a **time-extended k -set choice branching representation (k -TSTCB)** if there exists a ϕ_k such that q satisfies (4).*

We will provide two theoretical results regarding response time: inferring memory level; revealing preference under limited data:

1. **Inferring memory level.** From **Proposition 1.**, CB with different memory level sees a nested connection regarding explanatory range in choice data space $I = \{(x, A) : A \in \mathcal{X}, x \in A\}$. Thus, a choice data p which lies inside k -CB can be also explained by $k+1$ -CB. Then a problem arises: p is generated by k -CB or $k+1$ -CB? Such problem is universal in stochastic choice theory. Explanatory range of one model usually intersects with the others. When we collect a choice data p that falls in the intersection (p can be explained by both Model 1 and Model 2), we cannot determine data is generated from which model. As an extra information, response time can be used to distinguish CB with different memory level. An key observation is the following: CB with lower memory level will have longer response time. This is because CB with lower memory level forgets will branch to repeated alternatives more frequently as it forgets the traversed alternatives.
2. **Revealed preference under limited data.** In real-world choice data, limitness can be generally classified into two categories. First, the sample size of choice observations from an assortment is too small. So that the choice frequencies are a wrong estimator of choice probabilities. Second, the offered assortments are limited. We can only collect choice data from some assortments $\mathcal{A} \subset \mathcal{X}$. In both cases, we exploit response time as a compensation to reveal preference. A key observation is: If a consumer prefers an alternative more, he will choose it with a shorter response time.

4.2 Revealed Preference with Response Time

We consider two cases: revealed preference under limited assortments and revealed preference under small choice observations. First, we give a deterministic example to show how response time is informative in revealed preference

Example 4. Let $X = \{a, b, c, d\}$, $A = \{a, b\}$. Consider two deterministic choice branching process

1. $a \rightarrow b \rightarrow c \rightarrow d$
2. $c \rightarrow d \rightarrow a \rightarrow b$

Both branching processes result in the choice of a . However, the former requires one unit of time, whereas the latter requires three units. This indicates that consumer I has a stronger preference for a and exhibits a faster response.

We start our analysis with 0-CB. As we discussed before, the branching rule of 0-CB is equivalent to the weight w in MNL. The following proposition showed the relation between response time and the weight of an alternative.

Proposition 4. In 0-CB, the average response time of choosing x from A $\tau(x, A)$ is strictly decreasing on $\psi(x)$ (or $\phi(x), w(x)$).

Proof. We calculate the average response time of choosing x from A for 0-CB ψ .

$$\begin{aligned} T &= \psi(x) \cdot t + \sum_{a \in X \setminus A} \psi(a) \psi(x) \cdot 2t + \sum_{a \in X \setminus A} \sum_{b \in X \setminus A} \psi(a) \psi(b) \psi(x) \cdot 3t + \dots \\ &= \psi(x) \cdot t \cdot (1 + 2 \sum_{a \in X \setminus A} \psi(a) + 3 \sum_{a \in X \setminus A} \sum_{b \in X \setminus A} \psi(a) \psi(b) + \dots) \end{aligned} \quad (5)$$

$$\begin{aligned} &\psi(x) \cdot t \cdot \left(1 + \sum_{a \in X \setminus A} \psi(a) + \sum_{a \in X \setminus A} \sum_{b \in X \setminus A} \psi(a) \psi(b) + \dots \right. \\ &\quad \left. + \sum_{a \in X \setminus A} \psi(a) \left(1 + 2 \sum_{a \in X \setminus A} \psi(a) + 3 \sum_{a \in X \setminus A} \sum_{b \in X \setminus A} \psi(a) \psi(b) + \dots \right) \right) \end{aligned} \quad (6)$$

Let S denotes $1 + \sum_{a \in X \setminus A} \psi(a) + \sum_{a \in X \setminus A} \sum_{b \in X \setminus A} \psi(a) \psi(b) + \dots$ and U denotes $1 +$

$2 \sum_{a \in X \setminus A} \psi(a) + 3 \sum_{a \in X \setminus A} \sum_{b \in X \setminus A} \psi(a) \psi(b) + \dots$. Then

$$(5) = \psi(x) \cdot t \cdot U$$

$$(4) = \psi(x) \cdot t \cdot (S + \sum_{a \in X \setminus A} \psi(a) \cdot U)$$

As we showed in the proof of **Proposition 2.**, $S = \frac{1}{\sum_{a \in A} \psi(a)}$. Then by $(5) = (6)$,

$$U = \frac{S}{\sum_{a \in A} \psi(a)} = \frac{1}{\sum_{a \in A} \psi(a) \cdot \sum_{a \in A} \psi(a)}.$$

Then

$$T = \psi(x) \cdot t \cdot U = \frac{\psi(x)}{\sum_{a \in A} \psi(a) \cdot \sum_{a \in A} \psi(a)} \cdot t. \quad (7)$$

By taking differentiation,

$$\frac{\delta T}{\delta \psi(x)} = \frac{-\sum_{a \in A} \psi(a) \sum_{b \in X \setminus A} \psi(b) - \psi(x)}{(\sum_{a \in A} \psi(a) \cdot \sum_{a \in A} \psi(a))^2} < 0.$$

Q.E.D.

Proposition 4 has the following implication. Consider two 0-CB consumers, ψ_1 and ψ_2 , and suppose that, after normalization, $\psi_1(x) > \psi_2(x)$. In finite samples, however, stochastic variation may obscure this underlying preference ordering. In particular, when the number of observations is small, it is possible to observe that ψ_2 chooses x from a given assortment A frequently as, or even more frequently than ψ_1 . In the extreme case where the assortment A is offered only once to each consumer and both choose x , choice data alone provide no basis for distinguishing which consumer has a stronger preference for x . Response time, however, offers an additional source of information. Under the model, consumers who place a higher preference on an alternative tend to locate it more quickly in the search process and therefore exhibit shorter response times. Consequently, differences in response times can help recover the underlying preference ranking even when choice data are uninformative.

Chapter Five: Estimation and Numerical Experiment

5.1 Maximum Likelihood Estimation

We formulate the estimation problem for choice model. First we specify the type of data that we can collect. A data $N \in \mathbb{Z}_{++}^{|I|}$ is tuple of natural numbers. $N(x, A)$ means the number of observation (x, A) in the choice data. We formulate the estimation of choice model as a maximum likelihood estimation. The likelihood function can be represented as follows

$$\ell(N|\mathbf{Q}) = \sum_{(x,A) \in I} N(x, A) \cdot \log(p_{Model}(x, A|\mathbf{q}))$$

where $\mathbf{q}$ are the parameters and $p_{Model}(x, A | \mathbf{q})$ is the stochastic choice rule for the choice model conditional on parameter $\mathbf{q}$. Our main goal is to find the optimal $\mathbf{Q}$ to maximize the log-likelihood function.

$$\begin{aligned} & \text{maximize: } \ell(N|\mathbf{Q}) \\ & \text{s. t. } \mathbf{q} \in \mathbf{Q} \end{aligned} \tag{8}$$

, where $\mathbf{Q}$ is the feasible parameter space. An equivalent formulation of MLE is to formulate it on p space and constrain p with explanatory range $\mathcal{D}_{Model}$.

$$\begin{aligned} & \text{maximize: } \sum_{(x,A) \in I} N(x, A) \cdot \log(p(x, A)) \\ & \text{s. t. } p \in \mathcal{D}_{Model} \end{aligned} \tag{9}$$

Formulation (8) and (9) have their advantages in different scenario. When the paramter dimension is larger that p dimension it is better to apply formulation (3). As an example, we provide the MLE formulation for MNL model and RUM.

Example 5. MNL estimation

$$\begin{aligned} & \text{maximize: } \sum_{(x,A) \in I} N(x, A) \cdot \left(\log(e^{w(x)}) - \log \left(\sum_{a \in A} e^{w(a)} \right) \right) \\ & \text{s. t. } \sum_{x \in X} e^{w(x)} = 1 \end{aligned}$$

Example 6. RUM estimation

$$\begin{aligned}
& \text{maximize:} && \sum_{(x,A) \in I} N(x,A) \cdot \log(p(x,A)) \\
& \text{s. t.} && \sum_{x \in A} p(x,A) = 1 && \forall A \in \mathcal{X} \\
& && p(x,A) \geq 0 && \forall A \in \mathcal{X} \forall x \in A \\
& && \sum_{B: A \subseteq B \subseteq X} (-1)^{|B|-|A|} p(x,B) \geq 0 && \forall A \in \mathcal{X} \forall x \in A
\end{aligned}$$

Note that the characterization of RUM explanatory range is given by Falmagne [11].

5.2 Numerical Experiment

Synthetic data generation. We consider a setting with four alternatives, $X = \{a, b, c, d\}$. The data are generated from a mixture of three MNL models. We adopt this specification to ensure that the resulting choice data cannot be perfectly rationalized by a single MNL model, while remaining consistent with the broader RUM framework. The parameters of the three MNL components are specified as follows:

$$\begin{aligned}
& \lambda_1 = 0.2, \lambda_2 = 0.3, \lambda_3 = 0.5; \\
& w_1(a) = 1, w_1(b) = 1, w_1(c) = 100, w_1(d) = 10; \\
& w_2(a) = 1, w_2(b) = 100, w_2(c) = 1, w_2(d) = 10; \\
& w_3(a) = 100, w_3(b) = 1, w_3(c) = 1, w_3(d) = 10.
\end{aligned}$$

For each of the 15 possible assortments, we simulate 50 independent choice observations drawn from the corresponding probability distribution.

$$p(x,A) = \lambda_1 \cdot \frac{w_1(x)}{\sum_{a \in A} w_1(a)} + \lambda_2 \cdot \frac{w_2(x)}{\sum_{a \in A} w_2(a)} + \lambda_3 \cdot \frac{w_3(x)}{\sum_{a \in A} w_3(a)}.$$

This procedure yields a choice dataset comprising 750 observations in total. We then formulate the maximum likelihood estimation (MLE) problem in `Julia JuMP` for several models, including the MNL, a two-component MNL mixture, RUM, and the CB model under different memory levels. The resulting optimization problems are solved using the `Gurobi` optimizer. The following are the log likelihood value of choice models on this data set.

Model	Log-likelihood
MNL	5321.8
RUM	4816.3
2-MNL	4937.2
1-CB	5002.2
2-SQCB	4988.3
2-STCB	5032.1

Table 2. results of MLE on synthetic data

The results indicate clear differences in model fit across specifications. The MNL model exhibits the highest log-likelihood value, suggesting the poorest fit among the candidates, reflecting its well-known limitation in capturing complex substitution patterns. In contrast, the RUM achieves the lowest log-likelihood value, indicating the best fit, which is consistent with its high degree of flexibility. Among intermediate models, the mixture model (2-MNL) improves substantially over the single MNL, highlighting the importance of accommodating heterogeneity. The Choice Branching (CB) models further improve the fit relative to 2-MNL, suggesting that incorporating sequential search structure provides additional explanatory power beyond mixture-based approaches. Within the CB family, models with richer structure perform better. In particular, the 2-STCB model achieves the lowest log-likelihood among CB variants, outperforming both 1-CB and 2-SQCB. This suggests that allowing for more flexible branching dynamics (and potentially richer memory or state dependence) enhances the model's ability to capture the underlying data-generating process.

Overall, the results support the view that the CB framework provides a useful middle ground between the tractability of simple models (such as MNL) and the flexibility of RUM, delivering improved empirical performance while maintaining structural interpretability.

Conclusion Remarks

This thesis investigates the limitations of classical stochastic choice models, such as the Multinomial Logit (MNL) and Random Utility Models (RUM), in capturing the sequential nature of consumer search and the constraints of human memory. By introducing the Choice Branching Model (CB), we provide a unified framework that endogenizes the search process and accounts for memory-dependent behavior.

The key theoretical contributions are as follows. We establish that the Choice Branching Model serves as a bridge between several foundational models. Specifically, we prove that the MNL model is a special case of CB with zero memory, the Markov Chain Model (MCM) corresponds to a memory level of one, and the RUM is equivalent to a CB model with full memory of the search set. Our analysis of k -sequence branching rules demonstrates a clear link between memory capacity and search dynamics. Higher memory levels allow consumers to avoid redundant searches, thereby optimizing the decision-making path. A significant hurdle in discrete choice analysis is the identification problem. We show that while choice data alone may not distinguish between competing search protocols, the inclusion of Response Time provides the necessary variation to identify parameters. We find that RT serves as a proxy for search depth, allowing for a more granular estimation of consumer preferences.

As a practical implication, our findings suggest that firms and digital platforms can significantly improve user experience by tailoring information environments to estimated memory capacities. For instance, in high-complexity environments where users exhibit "low-memory" search patterns, reducing the length of consideration sets or increasing the salience of core options can reduce search friction and improve welfare.

While this research provides a robust foundation for memory-based choice branching, several avenues remain open for future exploration.

Endogenous Attention and Search Costs. Future models could integrate *rational inattention* theory into the CB framework. Rather than assuming exogenous branching rules, one could model how consumers strategically allocate attention based on the perceived cost of processing information versus the expected utility of the search.

Empirical Validation with Granular Data. While our numerical experiments validate the model's consistency, applying the CB framework to large-scale clickstream data or eye-tracking experiments is a crucial next step. Testing the model's predictive power in high-

dimensional choice sets (e.g., e-commerce) would demonstrate its practical scalability.

Decomposing Response Time. We treat RT primarily as a function of the number of search steps. Future research could decompose RT into two distinct components: the time taken to "move" between options and the "processing time" required to evaluate each option's attributes. This would allow for a finer distinction between search efficiency and cognitive difficulty.

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附录 1：毕业设计（论文）中人工智能工具使用说明

依据《上海交通大学关于在教育教学中使用 AI 的规范》的要求，为充分发挥“AI+HI（人工智能+人类智慧，Artificial Intelligence + Human Intelligence）”在毕业设计（论文）中的作用，特制定《毕业设计（论文）中人工智能工具使用说明》。该说明旨在支持人工智能工具在毕业设计（论文）中发挥辅助与启发功能，同时通过合理披露使用方式，确保毕业设计（论文）的完成过程中能够充分发挥“人类智慧”的作用，并符合科研诚信的要求。

毕业设计（论文）标题：Rational Choice Models: Characterization and Estimation

学生姓名：程晟 学号：522120910051 指导教师：何滔滔

一、本文使用的 AI 工具清单

序号 ^a	AI 工具名称	版本/型号	毕业设计（论文）中应用部分 ^b	人为判断过程 ^c 说明
1	ChatGPT	5.0	D3	润色论文语言，根据章节一的内容生成中英文摘要
2	ChatGPT	5.0	C2	辅助写 Julia JuMP 的参数估计优化问题代码

注释：

^a可以根据实际情况增删行数；

^b可以对照“应用部分对照表”填写对应的编号表，若超过编号表内容，请根据实际填写；

“人为判断过程指的是依据何种方式判断 AI 工具生成的内容的准确性。

毕业设计（论文）中应用部分对照表

A. 研究准备阶段	A1. 选题与方向设定；A2. 文献检索与筛选
B. 研究设计阶段	B1. 研究思路设计；B2. 方法论构建；B3. 研究工具开发
C. 研究实施阶段	C1. 实验/数据分析；C2. 代码编写与调试；C3. 数学建模与计算
D. 论文写作阶段	D1. 大纲与逻辑优化；D2. 论文正文写作；D3. 语言与语法润色
E. 其他辅助环节	E1. 参考文献与格式；E2. 图表/多媒体生成；E3. 学术反馈与模拟答辩

二、使用说明

1. D3：ChatGPT 润色论文语言，根据章节一的内容生成中英文摘要；2. C2：ChatGPT 辅助写 Julia JuMP 的参数估计优化问题代码

附录 2：学术诚信声明

本人承诺：论文核心观点、实验数据及结论均独立完成，AI 工具仅用于辅助性工作，已明确说明所有生成内容并在文中对应位置进行脚注标注。

承诺人：程晟

日期：2026 年 5 月 24 日

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